

Area or Perimeter? Fix the Mix-Up

Answer Key

Grade 3

Quick Answers

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|--------------------------------|---|
| 1. 28 cm^2 | 5. ribbon length = 36 cm |
| 2. 26 m | 6. 15 square units |
| 3. (b) = 18 ft^2 , — | 7. (a) = 24 m, — |
| 4. (a) = 45 cm^2 , — | 8. (a) = 30 m^2 , (b) = 26 m, (c) = > |
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What is verified

5 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 4, 7. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.MD.C.7 – problems 1, 3, 4, 6, 8

3.MD.D.8 – problems 2, 5, 7, 8

Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

4. *If they got 28: added the four sides (found the perimeter) instead of multiplying length by width*
7. *If they got 36: multiplied side by side (found the area) instead of adding the four equal sides*
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- 1.** A rectangular sticker is 7 cm long and 4 cm wide. What is the area?

Solution: Area is the space covered inside, so multiply length by width: $7 \times 4 = 28$.

The area is

28 cm^2

- 2.** A garden is 8 m long and 5 m wide. How many meters of fence go all the way around it (the perimeter)?

Solution: Perimeter is the distance around, so add all four sides: $8 + 5 + 8 + 5$, which is $2 \times (8 + 5) = 26$. The fence is

26 m

- 3.** A rug is 6 ft long and 3 ft wide. (a) Circle the word that tells how much floor the rug covers. (b) Find that amount.

Model answer (a), instructor judges: AREA must be circled — “how much floor the rug covers” is the space inside the rug’s edges, not the distance around them.

Solution (b): Area is length times width: $6 \times 3 = 18$. The rug covers 18 ft^2

4. Find and fix. Mateo’s poster is 9 cm long and 5 cm wide. He writes: “The area is 28 square centimeters, because $9 + 5 + 9 + 5 = 28$.” (a) What is the real area? (b) Name Mateo’s error.

Solution (a): Area multiplies the side lengths: $9 \times 5 = 45$. The real area is 45 cm^2

Model answer (b), instructor judges: Mateo added the four sides, which gives the *perimeter* (the distance around the poster), not the area. Area counts the square centimeters inside, so it comes from length $\times$ width.

5. Nia glues a ribbon around the edge of a card 12 cm long and 6 cm wide. How many centimeters of ribbon does she need?

Solution: “Around the edge” means perimeter, so add the sides: $2 \times (12 + 6) = 36$. Nia needs 36 cm of ribbon.

6. On the grid, each little square is one square unit. The rectangle is 5 units long and 3 units wide. What is its area?

Solution: Count the grid squares or multiply: 3 rows of 5 squares is $5 \times 3 = 15$. The area is 15 square units

7. Find and fix. Jaya’s square garden has 6 m sides. She works out $6 \times 6 = 36$ and buys 36 m of fence. (a) How many meters of fence does the garden actually need? (b) Explain what Jaya mixed up.

Solution (a): Fencing goes *around* the square, so add the four equal sides: $4 \times 6 = 24$. The garden needs 24 m of fence.

Model answer (b), instructor judges: $6 \times 6 = 36$ is the *area* of the garden — the square meters of ground inside. Fence length is the perimeter, the distance around the edge, so Jaya should have added the four 6 m sides instead of multiplying them.

8. Challenge. A play area is 10 m long and 3 m wide. (a) How many square meters of grass cover it? (b) How many meters of fence go around it? (c) Compare the two numbers with $>$, $<$, or $=$.

Solution (a): Area: $10 \times 3 = 30$, so the grass covers 30 m^2

Solution (b): Perimeter: $2 \times (10 + 3) = 26$, so the fence is 26 m

Solution (c): Compare the two numbers: 30 is more than 26, so $30 > 26$

— for this rectangle the area number is bigger than the perimeter number. (Careful: for other rectangles the perimeter number can be the bigger one!)